

# NDFU — Results Report #1: GSS vs GSS-DP vs GSS-SH

**Simulations run:** 2026-06-19 (GSS-SH diffusion sweep) **Analysis run:** 2026-06-21 (regression, phase transition, plots) **Report written:** 2026-06-21

This report consolidates the new results from adding the **attribute-shuffle null model (GSS-SH)** as a third topology alongside the plausible homophilic networks (GSS) and the degree-preserving randomized baseline (GSS-DP). It is a standalone record; the main manuscript is NOT yet modified.

## 1. The three topologies (what each one is, and what it destroys)

| Label         | Construction                                                                       | Degree           | Clustering / triangles / modularity | Homophily (attribute↔position) |
|---------------|------------------------------------------------------------------------------------|------------------|-------------------------------------|--------------------------------|
| <b>GSS</b>    | ERGM-imputed plausible networks (empirical homophily)                              | —                | preserved                           | <b>present</b>                 |
| <b>GSS-DP</b> | degree-preserving rewiring ( <b>rewire</b> , <b>keeping_degseq</b> , niter = 20·m) | <b>preserved</b> | <b>destroyed</b>                    | <b>destroyed</b>               |
| <b>GSS-SH</b> | topology byte-identical; node attributes randomly <b>permuted</b>                  | <b>preserved</b> | <b>preserved (identical)</b>        | <b>destroyed</b>               |

The logic of the design: **GSS-DP destroys topology and homophily simultaneously**, so it cannot say which one drives the adoption premium. **GSS-SH destroys only homophily** (the topology is literally the same graph, with attributes shuffled across nodes), so the contrast **GSS-DP vs GSS-SH isolates the contribution of the higher-order topology**, and **GSS-SH vs GSS isolates the contribution of homophily**.

### 1.1 Network-statistics verification (GSS vs GSS-SH; 100 vs 100 networks)

Confirms the shuffle does exactly what it should: topology identical, homophily annihilated.

| Family        | Statistic            | GSS     | GSS-SH  | Status           |
|---------------|----------------------|---------|---------|------------------|
| B (topology)  | global clustering    | 0.06475 | 0.06475 | <b>identical</b> |
| B (topology)  | triangles            | 11,200  | 11,200  | <b>identical</b> |
| B (topology)  | modularity (Louvain) | 0.2903  | 0.2903  | <b>identical</b> |
| B (topology)  | degree assortativity | 0.2907  | 0.2907  | <b>identical</b> |
| B (topology)  | mean degree          | 28.93   | 28.93   | <b>identical</b> |
| C (homophily) | assortativity, race  | 0.441   | −0.001  | collapsed → 0    |

| Family        | Statistic                                             | GSS   | GSS-SH  | Status        |
|---------------|-------------------------------------------------------|-------|---------|---------------|
| C (homophily) | assortativity, age                                    | 0.463 | -0.001  | collapsed → 0 |
| C (homophily) | assortativity, religion                               | 0.418 | -0.0001 | collapsed → 0 |
| C (homophily) | mean social distance on ties (Gower d <sub>ij</sub> ) | 0.229 | 0.399   | <b>+74%</b>   |

(For reference, GSS-DP also collapses C to ~0, but additionally drops B: clustering -31%, triangles -31%, modularity -47%.)

## 2. Result A — The structural premium, DECOMPOSED (BAM regression)

**Model:** Big Additive Model (mgcv **bam**, binomial/logit, fREML, discrete), pooled across both contagion types, all seedings, all threshold parameters. **Specification:**  $\text{adopt} \sim s(\text{IUL}, \text{MSP}, k=10, \text{by}=\text{contagion\_type}) + \text{tau\_mean} + \text{tau\_sd} + \text{seed\_type} + \text{network\_type}$  **Reference levels:**  $\text{network\_type}$  = Empirical (GSS);  $\text{seed\_type}$  = random. **N rows:** Empirical 2,046,720 · ER 2,046,720 · SH 1,023,360 (SH exists for CollectiveAction only — there is no ATP-SH run). All coefficients below have  $p < 2 \times 10^{-300}$  (z given).

### 2.1 Topology coefficients (the core result)

| Outcome                | $\beta(\text{GSS-DP})$ | OR    | drop   | $\beta(\text{GSS-SH})$ | OR    | drop   | Deviance expl. |
|------------------------|------------------------|-------|--------|------------------------|-------|--------|----------------|
| <b>Total adoption</b>  | <b>-0.8398</b>         | 0.432 | -56.8% | <b>-0.7663</b>         | 0.465 | -53.5% | 82.6%          |
| <b>Social adoption</b> | -0.4321                | 0.649 | -35.1% | -0.4033                | 0.668 | -33.2% | 72.0%          |
| Rational adoption      | -0.1136                | 0.893 | -10.7% | -0.1934                | 0.824 | -17.6% | 97.1%          |

### 2.2 Decomposition of the total-adoption penalty (-0.8398)

| Component                                                                 | Estimate (log-odds) | Share of DP penalty |
|---------------------------------------------------------------------------|---------------------|---------------------|
| <b>Homophily</b> (= $\beta_{\text{SH}}$ )                                 | <b>-0.766</b>       | <b>91.3%</b>        |
| <b>Higher-order topology</b> (= $\beta_{\text{DP}} - \beta_{\text{SH}}$ ) | -0.073              | 8.7%                |

**Reading.** Destroying *only* homophily — keeping clustering, triangles, modularity, degree and the giant component byte-identical — already reproduces **91% of the adoption penalty** that the full randomization produces. The structural premium is **overwhelmingly a homophily effect, not a clustering effect**. The same network, the same individuals, the same ties; merely reassigning *who sits where* destroys nearly the whole premium.

The social channel tells the same story ( $\beta_{\text{SH}} = -0.403$  vs  $\beta_{\text{DP}} = -0.432$ ; homophily  $\approx 93\%$  of the social penalty), which is expected because the selective-influence filter  $\sigma(\text{MSP} - d_{ij})$  is exactly the part of the model that homophily feeds.

## 2.3 Full parametric tables (CollectiveAction + Innovation pooled)

### Total adoption

| Term                    | Estimate       | Std. Error | z      |
|-------------------------|----------------|------------|--------|
| (Intercept)             | 6.3837         | 0.01528    | 417.7  |
| tau_mean                | -6.7839        | 0.01486    | -456.5 |
| tau_sd                  | 5.8197         | 0.03469    | 167.7  |
| seed: central           | 0.1302         | 0.00486    | 26.8   |
| seed: closeness         | 0.1296         | 0.00486    | 26.7   |
| seed: eigen             | 0.1290         | 0.00486    | 26.6   |
| seed: marginal          | -0.0729        | 0.00483    | -15.1  |
| <b>network: ER (DP)</b> | <b>-0.8398</b> | 0.00351    | -239.0 |
| <b>network: SH</b>      | <b>-0.7663</b> | 0.00460    | -166.7 |

Smooths  $s(\text{IUL}, \text{MSP})$ : edf  $\approx 9.0$  for both contagion types,  $\chi^2 = 376,793$  (Innovation) / 588,300 (CollectiveAction),  $p \approx 0$ .

### Social adoption

| Term                    | Estimate       | Std. Error | z      |
|-------------------------|----------------|------------|--------|
| (Intercept)             | -0.3004        | 0.00771    | -39.0  |
| tau_mean                | -5.2524        | 0.01245    | -421.8 |
| tau_sd                  | 4.3171         | 0.02992    | 144.3  |
| seed: central           | 0.0370         | 0.00421    | 8.8    |
| seed: closeness         | 0.0379         | 0.00421    | 9.0    |
| seed: eigen             | 0.0365         | 0.00421    | 8.7    |
| seed: marginal          | -0.0215        | 0.00423    | -5.1   |
| <b>network: ER (DP)</b> | <b>-0.4321</b> | 0.00297    | -145.5 |
| <b>network: SH</b>      | <b>-0.4033</b> | 0.00390    | -103.4 |

### Rational adoption

| Term        | Estimate | Std. Error | z          |
|-------------|----------|------------|------------|
| (Intercept) | 0.4277   | 0.00679    | 62.9       |
| tau_mean    | -0.1931  | 0.01059    | -18.2      |
| tau_sd      | 0.0393   | 0.02647    | 1.5 (n.s.) |

| Term                    | Estimate       | Std. Error | z     |
|-------------------------|----------------|------------|-------|
| seed: central           | −0.0582        | 0.00374    | −15.6 |
| seed: closeness         | −0.0558        | 0.00374    | −14.9 |
| seed: eigen             | −0.0555        | 0.00374    | −14.8 |
| seed: marginal          | 0.0278         | 0.00375    | 7.4   |
| <b>network: ER (DP)</b> | <b>−0.1136</b> | 0.00262    | −43.3 |
| <b>network: SH</b>      | <b>−0.1934</b> | 0.00340    | −56.8 |

**Note on the rational channel (caveat).** Rational adoption is  $q_i \leq \Gamma$ , which is node-local and uses a  $q_i$  distribution identical across all three topologies, so the *direct* topology effect should be  $\approx 0$ . The coefficients are small (OR 0.82–0.89) but non-zero, and unusually the SH penalty (−0.193) exceeds the DP penalty (−0.114). This is almost certainly a **second-order seeding/accounting interaction**: position-based seeding (central/closeness/eigenvector) and the asynchronous update change how adoptions get tallied as rational vs social near the IUL boundary when the social dynamics shift. It does not affect the Total/Social conclusions and should be a one-line footnote in the SM, not a headline.

### 3. Result B — Universality / criticality, across the three topologies

**Method:** susceptibility power-law fit (Method 4 of [07\\_phase\\_transition\\_deeper.R](#)): for each fixed IUL, find MSP\_c (steepest adoption jump) and fit  $\chi = \text{Var}(\Phi) \sim |\text{MSP} - \text{MSP}_c|^{(-\gamma)}$  on the super-critical side. Seeding = random,  $\tau_{\text{mean}} = 0.4$ ,  $\tau_{\text{sd}} = 0.12$ .

| IUL         | GSS — $\gamma$ (MSP_c)           | GSS-DP — $\gamma$ (MSP_c)        | GSS-SH — $\gamma$ (MSP_c)        |
|-------------|----------------------------------|----------------------------------|----------------------------------|
| 0.200       | 0.977 (0.250)                    | 3.709 (0.417)                    | 0.831 (0.417)                    |
| 0.225       | 0.892 (0.250)                    | 3.667 (0.417)                    | 0.751 (0.417)                    |
| 0.250       | 0.892 (0.250)                    | 3.667 (0.417)                    | 0.751 (0.417)                    |
| 0.275       | 0.873 (0.250)                    | 4.667 (0.333)                    | 1.977 (0.333)                    |
| <b>mean</b> | <b><math>\approx 0.91</math></b> | <b><math>\approx 3.93</math></b> | <b><math>\approx 1.08</math></b> |

**Reading. GSS-SH keeps the mean-field universality ( $\gamma \approx 1$ ); GSS-DP does not ( $\gamma \approx 4$ ).** Since GSS-SH preserves the higher-order topology and destroys only homophily, while GSS-DP destroys the topology:

→ **the continuous, universal ( $\gamma \approx 1$ ) character of the transition is anchored by the TOPOLOGY (clustering / community structure), NOT by homophily.** Removing homophily alone leaves the transition continuous and mean-field; only removing the higher-order topology collapses it into the discontinuous  $\gamma \approx 4$  regime.

**Critical openness shifts.** GSS tips at  $\text{MSP}_c = 0.25$ ; *both* GSS-DP and GSS-SH tip at  $\text{MSP}_c = 0.417$ . So removing homophily raises the social porosity needed to ignite a cascade — but the *order* of the transition (continuous vs discontinuous) is set by topology, independent of that shift.

**Caveats.** (i) The GSS-DP  $\gamma \approx 4$  is not a genuine power-law exponent — a first-order discontinuity has no diverging-power-law susceptibility, so that fit is a fingerprint of "not a power law", not a real  $\gamma$ . The meaningful contrast is that GSS-SH gives a *clean*  $\gamma \approx 1$  at the *same* MSP\_c (0.417) where GSS-DP fails. (ii) IUL = 0.275 is the noisy fitting point for all three (GSS 0.87, SH 1.98); the means are carried by the three stable points. (iii) coarse 13-level MSP grid — the usual finite-grid fragility.

## 4. The headline: a clean DOUBLE DISSOCIATION

| Topology      | Destroys homophily? | Destroys topology? | Volume premium ( $\beta_{\text{total}}$ ) | Universality ( $\gamma$ )            |
|---------------|---------------------|--------------------|-------------------------------------------|--------------------------------------|
| GSS           | no                  | no                 | — (reference)                             | $\approx 1$ (continuous)             |
| <b>GSS-SH</b> | <b>yes</b>          | no                 | <b>-0.766</b> ( $\approx 91\%$ of full)   | <b><math>\approx 1</math> (kept)</b> |
| <b>GSS-DP</b> | yes                 | <b>yes</b>         | -0.840                                    | <b><math>\approx 4</math> (lost)</b> |

### Two structural properties do two different jobs:

- **Homophily → VOLUME.** It governs *how much* spreads (the structural premium). GSS-SH, which destroys only homophily, already kills  $\sim 91\%$  of it.
- **Topology / clustering → CONTINUITY.** It governs *how* the system tips — smoothly (continuous, universal) vs abruptly (discontinuous). GSS-SH keeps  $\gamma \approx 1$ ; only GSS-DP, which destroys the topology, loses it.

This is a genuine double dissociation: each manipulation knocks out one outcome while sparing the other (SH kills the premium but spares universality; the topology destruction in DP is what kills universality). It cleanly separates the project's two headline results and assigns them to *distinct, identifiable structural causes* — turning two correlations into two mechanisms.

## 5. Provenance (files)

- **Diffusion results (SH):** `output/04_GSS_SH_diffusion_sims/` (25 RDS, 5 seedings  $\times$  all sd/mean)
- **Regression models (3-level network\_type):**  
`output/06_regression_analysis/models/bam_{total,social,rational}{,_counts}.rds + _summary.txt`
- **Critical exponents (SH):** `output/07_phase_transition/GSS_SH/gamma_exponents.csv`; plots in `plots/07_phase_transition/GSS_SH/`
- **Adoption heatmaps (SH):** `plots/04_GSS_SH_diffusion_sims/` (20 PDFs)
- **Network construction & verification:** `playground/checks-claude/06_make_shuffle_networks.R`, `data/02_GSS_SH_network/` (100 nets), `playground/checks-claude/data/shuffle_verification.csv`
- **Diagnosis (GSS vs DP):** `playground/checks-claude/05_structural_diagnosis.R`, `STRUCTURAL_DIAGNOSIS.md`
- **Adapted scripts:** `scripts/06_regression_analysis.R` (3-level), `scripts/05_GSS_diffusion_plot_main.R` (SH type), `scripts/07_phase_transition_deeper.R` (NDFU\_TOPO env var); SH topology branch in

```
scripts/04_GSS_diffusion_sims.R; runner  
scripts/04_GSS_SH_diffusion_sims_main.R
```

## 6. Open questions / next

- **$\beta$  extraction and the Widom relation** still need a denser near-MSP\_c grid before  $\gamma/\beta/\delta$  can be reported as a rigorous universality result.
- **Why does removing homophily preserve universality?** Worth a short mechanistic argument: clustering provides the local "averaging cells" that produce mean-field coupling, independent of whether neighbors share attributes.
- **Rational-channel second-order effect** (§2.3) — confirm the seeding/accounting explanation if it ever needs to go in the SM.